



Agile Methodologies: Organizational Adoption Motives, Tailoring, and Performance

John F. Tripp & Deborah J. Armstrong

To cite this article: John F. Tripp & Deborah J. Armstrong (2018) Agile Methodologies: Organizational Adoption Motives, Tailoring, and Performance, Journal of Computer Information Systems, 58:2, 170-179, DOI: [10.1080/08874417.2016.1220240](https://doi.org/10.1080/08874417.2016.1220240)

To link to this article: <https://doi.org/10.1080/08874417.2016.1220240>



Published online: 04 Oct 2016.



Submit your article to this journal [↗](#)



Article views: 3358



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 21 View citing articles [↗](#)



Agile Methodologies: Organizational Adoption Motives, Tailoring, and Performance

John F. Tripp^a and Deborah J. Armstrong^b

^aBaylor University, Waco, TX, USA; ^bFlorida State University, Tallahassee, FL, USA

ABSTRACT

Today, organizations tailor the practices from several agile methodologies to serve their particular environment. But are there situations that drive how an organization should tailor methodologies in a particular manner? This article places 12 commonly used agile development practices into a typology based upon whether they are primarily project management focused or software development approach focused and examines how organizations' motivations for adopting agile impact the practices they adopt. Finally, it explores how a fit between an organization's motives for agile method adoption and the tailored agile practices it adopts may lead (or not lead) to differences in project performance.

KEYWORDS

Agile methodologies; IT project management; software development; adoption motivation; method tailoring; project performance

Introduction

Organizations have various motivations when adopting a software development method. Within the information systems development domain, a software development method (also referred to as a methodology, process, or approach [12]) is a prescribed set of related, often interdependent practices, which is formulated with the intent of improved planning and execution of the software development process. Agile software development methodologies (henceforth, agile methodologies) approach the software development process using practices that allow software requirements and solutions to evolve through collaboration within self-organizing, cross-functional teams that work in short cycles (i.e., sprints) to facilitate rapid innovation [9]. Agile methodologies have become well-accepted, with over 65% of companies reporting some type of use of agile methodologies for their software development projects [3].

Agile methodologies have become widely popular since the publication of the agile manifesto in 2001 [39]. While various methodologies (e.g., extreme programming (XP) [2], scrum [33], feature-driven development (FDD) [29], and lean software development (LSD)) [30] have claimed the designation of “agile,” they advocate significantly different sets of agile practices. Agile practices are software development tasks or activities that are used to implement the agile method's principles and values. Differences between the practices defined in each method are to be expected, as each method emerged from a different context, with somewhat different goals. For example, scrum [33] is an agile method that focuses primarily on managing project team tasks through practices such as a daily standup meeting, iteration planning, and delivery in short sprints.¹ In contrast, XP [2] is an agile method that advocates practices that are focused on quality and software engineering techniques (e.g., pair programming, unit testing).

Agile practitioners view the various practices defined by the broad family of agile methodologies as a “toolkit” that can be drawn upon and configured as necessary [39]. Selecting practices from multiple agile methodologies is called agile method tailoring [7]. Because of this, even when organizations adopt the same agile method(s), there can be wide variation in the agile practices adopted. For example, say two organizations—FoodCo and BevCo—claim to use the XP agile method. FoodCo uses the pair programming, planning game, and iteration planning practices, whereas BevCo uses coding standards, pair programming, and test-driven development practices. Both organizations report that they are using the same agile method, but are they? And are they getting the same benefits from the practices adopted? While Campanelli and Parreiras [5] recently published a systematic literature review of agile method tailoring approaches, little research has been conducted that investigates the variation in the manner in which agile methodologies are adopted and how this variation may lead to different outcomes (for exception see [13]).

While several studies have looked at success factors in the adoption and use of agile in general [6, 25, 27], and challenges in the adoption of agile methodologies [28], little empirical evidence has emerged addressing the factors that drive the adoption of different agile practices, and whether differences in the set of agile practices adopted impact the performance of agile teams. Further, we are not aware of research that has investigated what factors drive different patterns of adoption (i.e., agile method tailoring), or how differences in agile method tailoring may lead to different outcomes.

In this article, we use two data sources to explore the phenomenon of agile method tailoring. First, we use VersionOne 2011 survey data, which is an annual survey administered by the VersionOne corporation [35]. In addition to this extremely large dataset, we conducted supplemental in-

CONTACT John F. Tripp ✉ john_tripp@baylor.edu Information Systems, Baylor University, Waco, TX 76798-7151, USA.

¹In agile software development, work is contained in a regular, repeatable work cycle, known as a sprint or iteration.

depth interviews with agile practitioners. Using these two data sources, we investigated the following questions: (1) How do differences in organizational motivations for the adoption of agile methodologies (i.e., agile method tailoring) align with the agile practices implemented by the organization and, (2) is there a potential “fit” between an organization’s motivations for adopting agile methodologies, the agile practices adopted, and agile project performance? While motivations for agile method adoption have been proposed [10, 12, 16], in this empirical research, the motivations emerged from the VersionOne State of Agile 2011 data and were triangulated with interview data from agile consultants/evangelists. Our findings suggest that organizational adoption motives do drive differences in the focus of practices implemented and may drive differences in project performance outcomes.

Agile software development method adoption

The adoption of agile methodologies has been documented in the literature, with multiple case studies focusing on the adoption of agile methodologies in general and providing “lessons learned” for researchers and practitioners [12, 36, 37]. For example, Fruhling and de Vreede [12] looked at operationalizing XP techniques for a web-based system and found that it aided communication and flexibility. In addition to a global view of agile adoption, a few studies have looked at the adoption of specific components of agile, such as user stories [24]; XP practices [22, 25]; rapid application development [4]; and agile requirements engineering [32]. Finally, some studies have looked at the integration of agile methodologies and other processes (often referring to these as “hybrid” approaches) such as product line engineering [17]; plan-based requirements prioritization [31]; documentation-driven methodologies [18]; lean methodologies [38]; service-oriented methodologies [20]; and most recently capability maturity [11, 14, 34].

Outcomes of agile method adoption have also begun to be explored, particularly factors that lead to project success [6, 23, 27]. For example, Maruping et al. [26] found that the level of requirements change and outcome control interact with agile practices to influence software project quality, while Wood et al. [40] found that customer planning positively influenced developer performance. However, a key gap in current research is exploring the link between organizational adoption motives and the agile practices implemented, and ultimately, the performance of agile teams and projects.

Organizational motivations for the adoption of agile methodologies

The source data from the 2011 State of Agile Survey was used for this research, and VersionOne, a maker of agile team coordination and management software, provided the data to the authors after scrubbing any personal information. The survey, which is administered annually, contained over 40 questions and included more than 6000 respondents. We filtered the data to 2304 respondents who were knowledgeable in agile software development (the respondent indicated that they and their organization had utilized agile methodologies/

practices on at least one project and for at least 6 months). Specifically, the number of individuals that started the survey was 6042, and the number that completed the survey was 4235. From this number, a total of 1931 responses were removed from the dataset: 407 responses were eliminated due to incomplete data; 720 responses were eliminated because the respondent indicated they had no or very little agile knowledge or agile experience; and 804 responses were eliminated because the respondent indicated that the organization had no teams, projects, or locations engaged in agile development. This process resulted in a sample that was more representative of adopters of agile development methodologies and not all software developers.

Survey respondents were asked questions about the organization’s motivation(s) for using agile methodologies, which agile practices were used in the respondent’s organization, and what challenges the organization was facing in their use of agile methodologies. Question types included open, dichotomous, Likert type scales, and multi-response. The SPSS v.20 software package was used to conduct the analyses. The individuals in the sample had an average of 2.97 years of agile experience, and their organization’s level of agile experience averaged 3.75 years. The percentage of projects using agile methodologies in the respondents’ organizations averaged 31.8%. Table 1 provides additional demographic details for the survey respondents and provides evidence that the sample included a wide range of individuals with regard to their functional department and role.

Our study was exploratory in nature, so we sought to triangulate the VersionOne survey data with qualitative data to ensure that we collected data from a variety of perspectives. The authors conducted semi-structured interviews with agile software development experts. Many of the interview contacts were identified as attendees of the Agile Alliance 2013 Conference (agile2013.agilealliance.org/). All interviews were conducted by phone, lasting from 45 min to 90 min, with both authors participating in each interview. The interviews were recorded and transcribed verbatim. The interviewees had an average of 27 years of software development experience and were currently employed either as agile coaches or as directors of development organizations that self-identified as using agile methodologies.

Table 1. Respondent profiles for VersionOne survey.

		Frequency	Percentage
Department	IT/Support	606	26.3
	Marketing/Sales	48	2.1
	Services	95	4.1
	Software Development	1379	59.9
	Other	176	7.6
Job Title/Team Role	CEO/COO/President	46	2.0
	CIO/CTO	67	2.9
	Consultant	186	8.1
	Developer	93	4.0
	Development Manager	302	13.1
	IT Staff	29	1.3
	Product Manager	127	5.5
	Project Manager	464	20.1
	QA/Tester	78	3.4
	Senior Developer	156	6.8
	System Architect	119	5.2
	Team Lead	213	9.2
	VP/Director of Development	222	9.6
	Other	202	8.8

Table 2. VersionOne survey adoption motive factor structure.

Adoption motive	M1	M2	M3
Enhance software quality	0.683		
Enhance software maintainability/extensibility	0.730		
Improved/increased engineering discipline	0.733		
Accelerate time-to-market		0.800	
Increase productivity		0.645	
Reduce cost		0.618	
Enhance ability to manage changing priorities			0.751
Improve alignment between IT and business objectives			0.684
M1 = Improve Software Quality; M2 = Improve Efficiency; M3 = Improve Effectiveness (no cross-loadings over 0.3).			

The factors were extracted based on eigenvalues (greater than 1) and resolved into three factors (). The five motives that were not included in further analysis did not reach the 0.6 item loading value on any factor [15].

Agile method adoption motives

Our first goal in this study was to investigate motives for the adoption of agile methodologies. Using the VersionOne survey data, we performed an exploratory factor analysis using principal component factor analysis and varimax rotation with kaiser normalization (scale 1: not important at all; 4: highest importance) of the responses to the question “How important were the following in your company’s decision to initially adopt agile development methodologies in your organization?” Thirteen motives were listed in the survey, eight of which loaded onto three factors. Table 2 illustrates the adoption motives and factor loadings. To triangulate this structure with another data source, we also asked interview participants “What drives companies to adopt agile methodologies?” While the wording of the answers did not match exactly, both the survey data and our interview responses were conceptually consistent. Consistent with the survey data, our qualitative analysis found that the motives clustered into three high-level categories which we label (1) improve software quality, (2) improve efficiency, and (3) improve effectiveness, which are detailed next.

The *Improve Software Quality* category consists of the adoption motives such as enhancing software quality, improving engineering discipline, and enhancing software maintainability. Our interview data also strongly supported improving software quality as a key motive for agile adoption. For instance:

Everybody says they want to *increase their quality*. — Agile Coach

Some want *higher quality*, some want faster to market, some want to be more responsive and more competitive. — Agile Coach

The business was always unhappy with the development executives because [...] we had *quality problems* with the product ... I was tired of being kicked in the butt all the time for delivering or inheriting software that was not put together well... — Development Executive

The *Improve Efficiency* category consists of adoption motives such as increasing productivity, accelerating time to market, and reducing costs. Further, our interview data supported improving efficiency as a key value proposition for the adoption of agile:

The company has a relatively new CTO that has come in and indicated that he’s going to have the organization be consistently agile throughout. Looking at kind of *increasing productivity* of the organization overall. — Agile Coach

A lot of them are picking up scrum management practices, and my experience is what they get out of that is a 20–30%, they get a little bit better customer visibility which is big kind of politically usually. They also get about a 20–30% *productivity enhancement*. — Agile Coach

The *Improve Effectiveness* category focuses on adoption motives such as enhancing the organization’s ability to manage changing priorities and improving the alignment between business objectives and IT. Our interview data supported improving effectiveness as a key motive for agile method adoption. For instance:

We got into the market the same year we started project. What we did in the business sense, holy crap, we got into market three months early and they made up their revenue for the entire year in a quarter. From that perspective that *business understood the value*. — Development Executive

A typology of agile practices

Before we identify how adoption motives affect variation in practice adoption (RQ1), we need to establish whether or not there is variation in the adoption of agile practices across organizations. To do this, we turned again to the VersionOne 2011 survey data. In this survey, respondents are asked to identify which of 25 different agile practices their organization has adopted. We determined the rank order of agile practices (i.e., the practices used by the largest number of respondents). While the survey asked respondents about their use of 25 agile practices (dichotomous scale), the adoption rate for the practices dropped significantly after the twelfth practice (52–41%). Further, we found that the average number of practices adopted was 11.5. Thus, we limited our analysis to the 12 most used agile practices.

Similar to our analysis of the adoption motives, we sought to identify any commonalities (i.e., categories) in the agile practices. To determine if there was any pattern or structure in the practices, we performed a two-round categorization exercise for the 12 practices. First, three expert agile developers from different organizations were asked to categorize the practices. These developers had between 5 and 14 years of agile development experience. At the conclusion of the initial categorization, two of the experts had developed two categories, which seemed to reflect a project management (PM) focus and a software development approach (SDA) focus. For these two experts, there was 95.9% agreement on the practice categorization. The third expert developed four categories, which we label discipline, management, metrics, and strategy. As PM includes the management of projects, tracking of metrics, and setting a strategy for completion, the authors merged these three categories into the PM category. The discipline category was congruent with the SDA category developed by the other two experts. When looking across the three experts and 12 practices, the final categorization scheme reflected a 97.2% level of agreement.

The two categories of practices that emerged were PM (which contained practices such as iteration planning) and software development approach (which contained practices such as unit testing). In the second round, two different agile developers conducted a card sort categorization of the

Table 3. Agile practices investigated.

Rank	Practice	Adoption rate	Practice category
1	Daily standup	86%	Project management
2	Iteration planning	81%	Project management
3	Unit testing	75%	Software development approach
4	Retrospectives	73%	Project management
5	Burndown	72%	Project management
6	Release planning	71%	Project management
7	Automated builds	61%	Software development approach
8	Velocity	61%	Project management
9	Continuous integration	61%	Software development approach
10	Coding standards	57%	Software development approach
11	Refactoring	54%	Software development approach
12	Test-driven development	52%	Software development approach

12 practices into the two categories. The overall level of agreement on the card sort categorization was just over 80.0%. This categorization scheme reflects our typology of agile practices, which is presented in Table 3. Before we proceed, we briefly describe the practices in each category.

Project management category

Six practices were identified for the PM practice category (Table 3, far right column). *Daily Stand Up* (Rank 1) refers to a meeting held each day for which every team member attends and provides information to the team regarding the work performed the previous day, the work planned for the day, and any blocking or coordination issues he or she has encountered.

Release Planning, *Iteration Planning*, and *Velocity* are each associated with the planning of work cycles. *Release Planning* (Rank 6) defines at a high level the order in which features will be deployed for a longer-term project timescale (several work cycles). *Iteration Planning* (Rank 2) is performed before each work cycle, as the team and customer together define the features included in the next work cycle, divide the features into tasks, and estimate the work to be performed. These practices build on the team's established *Velocity* (Rank 8), which defines a baseline amount of work that can be accomplished in a work cycle by a team.

Burndown (Rank 5) refers to the practice of visually tracking the progress of each work cycle with a graph that represents the amount of work that should be completed by a certain date and the amount of work that has been completed. Finally, *Retrospectives* (Rank 4) refers to the team meeting after each work cycle in which team members reflect on the positive and negative aspects of the previous work cycle and take corrective actions. In sum, the agile practices in the PM category focus on planning, coordination, work metrics, and communication to facilitate the software development process.

Software development approach category

Six practices were identified for the SDA category (Table 3, far right column). *Unit testing* (Rank 3) refers to the use of

dedicated test code that can be run (usually automatically) to test the effects of changes made to the system. *Automated Builds* (Rank 7) refer to the use of a code script to rebuild the software. This ensures that all developers have a baseline set of code before making changes. *Continuous Integration* (Rank 9) is, in essence, a combination of unit testing and automated builds in which teams often utilize a non-developer machine and the build script to rebuild the software product on a regular basis (sometimes after every change).

Coding Standards (Rank 10) refers to a set of norms regarding code-naming and consistency. By consistently formatting and structuring code, functionality is more consistently represented to multiple developers and can lead to developers' increased understanding of the appropriate way to perform code changes. *Refactoring* (Rank 11) refers to practices that lead to the removal of redundancy, elimination of unused functionality, and refresh obsolete designs.

Finally, *Test-Driven Development* (Rank 12) refers to the practice of writing test code before system code. This practice can lead to system code that is structured appropriately for testing. In sum, practices in the SDA category focus on coding, functionality, and testing to facilitate the software development process.

We next explored the relationship between the organizational motives for the adoption of agile methodologies and the tailoring of the agile practices in use to address RQ1.

Tailoring of agile methodologies

Recall that in software development, "agile method tailoring" is the process of customizing the agile method to meet the context and circumstances of use [7]. As one agile coach stated referring to agile methodologies,

"There is no book per se. It is always tailored to the organization. We start and the pieces aren't known, there's some scrum in it, there's some XP in it, there may be some FDD in it, there may be some Kanban lean software development. It really depends on the organization." — Agile Coach

Organizations might pursue agile methodologies because they are doing well in their software development efforts and want to improve in the sense of continuous improvement. On the other hand, organizations might pursue agile methodologies because of perceived weaknesses in their current software development efforts. Organizations' strengths or weaknesses in their previous software delivery outcomes might influence motivations to adopt agile methodologies. To explore the relationships/patterns between the agile adoption motive and the specific agile practices adopted, a correlation analysis was conducted.

We explored the relationship between the motivation factors and the 12 agile practices using Polychoric correlation, which estimates a correlation of a theoretically normally distributed continuous latent variables, which are measured ordinally [19]. Polychoric correlation is appropriate for our data as the response options in the VersionOne survey were dichotomous (1 = practice was used vs. 0 = practice was not used), rather than a scale that defined the extent of use. Significant correlations were identified between the three

motivation factors (i.e., categories) and the 12 agile practices. The data in Table 4 illustrate that the Improve Software Quality motive was negatively correlated with the PM agile practices and positively correlated with the SDA agile practices. The Improve Efficiency motive was positively correlated with the PM agile practices, but was uncorrelated with the SDA agile practices. The Improve Effectiveness motive was positively correlated with the PM agile practices and positively correlated with the SDA agile practice. It is important to note that the correlation coefficients' relatively low levels are due to the fact that the VersionOne survey records the use of the practice as a simple binary response. So while the levels are relatively low, the key takeaway is the pattern of significance illustrated in Table 4.

To further test the categories, we summed the practices by category (PM versus SDA) and conducted further correlation analysis. As illustrated in Table 5, we found that the pattern of adoption of agile practices is significantly different for the three adoption motivations. When the adoption of agile methodologies is primarily driven by a desire to improve software quality, there is a significant positive correlation between the adoption motive and the number of SDA-focused agile practices used by the organization ($0.134, p < 0.001$) and a significant negative correlation with the number of PM-focused agile practices used by the organization ($-0.074, p < 0.001$). When the initial agile adoption motivation is driven by a desire to increase efficiency, we see a significant positive correlation between the adoption motive and the number of PM-focused practices adopted by the organization ($0.127, p < 0.001$), while there is no correlation with

the number of software development practices adopted ($-0.014, ns$). Finally, for organizations with the initial agile adoption motivation of greater effectiveness, there is a significant positive correlation between the adoption motive and both categories of practices—i.e., organizations that pursue effectiveness tend to also adopt higher levels of both categories of practices (PM = $0.096, p < 0.001$; SDA = $0.053, p < 0.001$).

Based on our results, we speculate that when an organization's motive for the adoption of agile methodologies is to enhance software quality, it is reasonable to expect that a higher than average number of practices that are focused on the software engineering process would be adopted. Similarly, when the goal of the organization is to improve efficiency, it is likely that the organization focus on the adoption of PM techniques that can have an effect on coordination, communication, and planning. Finally, as the motive of effectiveness entails delivering a high-quality product, and doing it in an efficient manner, a broader pattern of adoption might be indicated. Thus, in answer to RQ1, we assert that the differences in the initial motives for the adoption of agile methodologies may drive the type of agile practices implemented by the organization. We use this finding to develop the concept of "fit" to address RQ2.

Achieving outcomes with tailored agile methodologies

To delve into the relationships between the motivations for adopting agile methodologies and the agile practices used by the organization, we sought to determine if a "fit" between the initial motivation for adopting agile methodologies and the agile practice adoption pattern was correlated with higher performance on a number of metrics. For example, if an organization's primary adoption motive was improved efficiency, a fit would exist for organizations that had adopted a higher than average number of PM-focused agile practices. Organizational theory suggests that performance outcomes should increase when a fit between the adoption motivation and agile practices used is present [21]. So, if the primary motivation for an organization using agile methodologies is to enhance software quality, and the development teams use a higher than average number of SDA-focused practices (i.e., motivation-practice fit), then the organization's outcomes of interest (e.g., project performance) should be higher than an organization with an enhance software quality motivation that uses a lower than average number of SDA-focused practices, (i.e., motivation-practice mis-fit).

To explore the potential fit, we divided the survey sample based upon the primary adoption motivation and then separated the organizations into "high" practice adoption and "low" practice adoption groups, by splitting at the mean. This provided four quadrants (Quadrant 3 = High PM–Low SDA, Quadrant 4 = High PM–High SDA, Quadrant 2 = Low PM–Low SDA, and Quadrant 1 = Low PM–High SDA (see Figure 1). The three instances of agile adoption motive–agile practice fit are indicated in the quadrant in Figure 1 (e.g., Quadrant 3—the Increase Efficiency motive with the High PM–Low SDA practice use creates a fit). An Analysis of Variance (ANOVA) was used to compare the means of each quadrant for each adoption

Table 4. Polychoric correlation analysis.

Agile practice	Adoption motive		
	Improve software quality	Improve efficiency	Improve effectiveness
Project management (PM) category			
Burndown	−0.090**	0.122**	0.025
Daily standup	−0.112**	0.087**	0.034
Iteration planning	−0.042	0.084**	0.091**
Release planning	0.003	0.155**	0.125**
Retrospectives	−0.065**	0.086**	0.061**
Velocity	−0.071**	0.077**	0.115**
Software development approach (SDA) category			
Automated builds	0.075**	−0.018	0.011
Coding standards	0.152**	−0.019	0.061*
Continuous integration	0.076**	−0.016	0.058*
Refactoring	0.108**	−0.047	0.064*
Test-driven development	0.188**	0.001	0.043
Unit testing	0.048*	0.045*	0.015

* Significance at 0.05 level (two-tailed).

** Significance at 0.01 level (two-tailed).

Table 5. Relationship of adoption motive to agile practice category adopted.

Initial adoption motivation	Agile practice category focus	
	Project management	Software development approach
Improve software quality	−0.074**	0.134**
Increase efficiency	0.127**	−0.014
Increase effectiveness	0.096**	0.053**

*Significance at 0.05 level (two-tailed).

**Significance at 0.01 level (two-tailed).

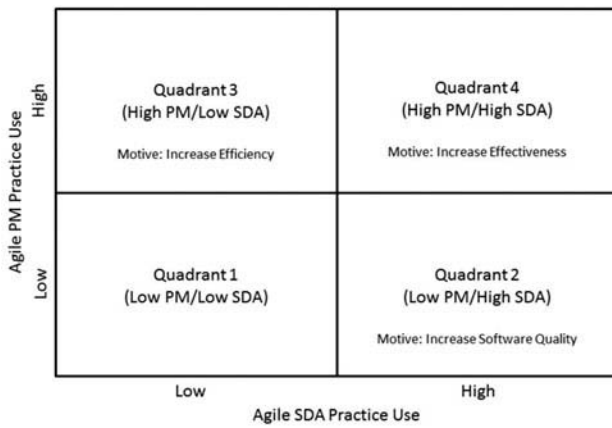


Figure 1. Adoption grouping quadrant definitions.

motivation on a variety of performance metrics. Significant differences in the number of PM or SDA practices for different adoption motives were found and are detailed next.

Looking at a number of outcome measures, we can see some interesting patterns that are shown in Table 6. To achieve the desired outcomes, it is important to focus on the intersection between the organization's motive for the adoption of agile methodologies and the measure of the desired performance outcome. For example, looking at the first row of Table 6, we see that the enhanced software quality outcome is associated with the use of a higher than average number of SDA-focused practices if the organization's initial adoption motivation is to Increase Software Quality (indicated by the "SDA" code in the column 2 cell). Thus, a code in a cell (e.g., PM, SDA, PM + SDA) suggests that higher performance in the outcome (e.g., increased project success %) is correlated with a higher than average number of practices in the indicated category (PM) for the particular organizational adoption motive (Improve Efficiency).

Note that in Table 6 when one category of practices (either SDA or PM) is associated with higher performance on a specific outcome, it does not imply that the adoption of agile practices from the other practice category is low. It simply means that the level of adoption of practices from the other practice category, whether high or low, does not correlate with the outcome. In addition, a cell value of "PM+SDA" (e.g., the cell at the intersection of the increased productivity outcome and the Increase Efficiency adoption motive) indicates that teams that had higher

productivity adopted a higher than average number of agile practices in both practice categories. Finally, the cell value of "PM or SDA" (e.g., the cell at the intersection of the improved response to changing priorities outcome and the Increase Efficiency adoption motive) indicates that teams that had higher performance on a specific outcome adopted a higher than average number of practices from one practice category. A dash in the cell (see the cell at the intersection of the enhanced software quality outcome and the Improve Effectiveness adoption motive) indicates that under this adoption motive, engaging in a higher than average number of PM- or SDA-focused practices has no link with the level of the performance outcome.

For example, if *Increasing Software Quality* (Table 6, column 2) is the adoption motive, for most outcome measures (e.g., increased project success percentage, improved engineering discipline, etc.) the organization would probably want to focus on utilizing a higher than average number of SDA-focused practices. However, if the outcome of interest is increased speed to completion and/or improved business-IT alignment, then incorporating both PM-focused practices and SDA-focused practices might be more beneficial. If the organization is looking to improve their response to changing priorities as an outcome, developers might want to focus on utilizing a higher than average number of PM-focused practices. Finally, if the organization is looking for increased productivity as an outcome, engaging in a higher than average number of SDA-focused practices and/or a high number of PM-focused practices seems to have no connection (as indicated by the dash in the cell at the intersection of the Improve Software Quality adoption motive and the increase productivity outcome). It appears that the outcome of increased productivity is outside of the influence of agile practices when the agile adoption motive is to Improve Software Quality. Each adoption motive pattern in Table 6 is summarized next.

To summarize, if the organizational motive for adopting agile methodologies is to Increase Software Quality we suggest that organizations might want to focus on using a higher than average number of SDA-focused agile practices, and depending on the desired outcome, organizations might also want to use a high number of PM-focused practices. If the organizational motive for adopting agile development is to Increase Efficiency its the opposite—increases in all of the outcomes seem to be consistent with engaging in a higher than average number of PM-focused practices and some outcomes seem to entail a higher than average number of SDA-focused practices

Table 6. Fit between adoption motivation and performance indicator.

Outcome/performance indicator	Motive 1: Increase SW quality	Motive 2: Improve efficiency	Motive 3: improve effectiveness
Enhanced software quality	SDA	PM	—
Improved engineering discipline	SDA	PM	—
Enhanced SW maintainability	SDA	PM	—
Increased speed to completion	PM + SDA	PM	—
Increased productivity	—	PM + SDA	—
Accelerated time-to-market	SDA	PM + SDA	PM
Improved response to changing priorities	PM	PM or SDA	PM
Improved IT and business alignment	PM + SDA	PM or SDA	—
Increased project success %	SDA	PM	SDA

PM = Project Management-focused practices; SDA = Software Development Approach-focused practices.

A code in a cell indicates that higher performance in the indicator is related to a higher than average number of practices in the indicated area for the particular organizational motive.

Average PM practices = 4.4, Average SDA practices = 3.5.

as well. It seems that to effect outcomes when the pursuit of increasing effectiveness is the motive for agile software development adoption requires efforts beyond the adoption of agile practices. Thus, if increasing the percentage of successful projects is the desired outcome, we recommend that organizations that wish to enhance software quality or increase effectiveness could focus effort on increasing the number of SDA-focused practices; whereas organizations that wish to enhance their process efficiencies could focus effort on increasing the number of PM-focused practices.

We assert that the results suggest that a fit between the organizational adoption motive and the adopted agile practices does co-occur with higher performance on a number of outcome measures. In related work, Cram [8] compared the values of project teams and their development approach and found that where alignment was high, perceptions of the systems development process was associated with satisfaction and enthusiasm; and where alignment was low, perceptions focused on frustration and discontent. Here fit refers to how well the agile practice used supports the adoption motive (and ultimately the outcome metric). We have found preliminary evidence indicating that a fit between the initial agile adoption motive and the practices incorporated into tailored agile methodologies has an association with project performance, thus addressing RQ2. We now provide more specific guidelines for organizations and project managers in their tailoring of agile methodologies.

Guidelines for tailoring agile methodologies

Recall that agile method tailoring “describes the overall process of selecting or adapting software practices” [1, p. 4]. For organizations seeking to implement agile methodologies, understanding the path of adoption that will likely lead to success is a complex task. Our findings suggest that there is not a simple formula that indicates which practices are the “right” practices in every situation. However, we have formulated four guidelines from this study that may be used to help organizations to develop a strategy for agile method tailoring. Three guidelines are focused on organization-level actions and one guideline is focused on team-level actions.

Organizations: Focus on the biggest pain, then expand adoption

As our Table 6 illustrates, the combination of initial adoption motive and outcome pursued determines the agile practices that if adopted are more likely to generate higher performance on the performance metric being sought. That being said, some organizations may have multiple adoption motives, and these may have conflicting practice categories that are indicated. For organizations in this situation, our interviews gave us additional insight into how organizations might want to focus their adoption efforts of agile methodologies. Several of our interview participants assert that organizations should focus on the most immediate pain point(s), and then expand adoption from there.

There’s a term in the consulting community called “scrum but”, and what it is, it’s an organization that says we’re doing scrum BUT here’s the things in scrum that we’re not doing that are in scrum. The reality most of the time, when you hear “scrum but” is they’re picking the things that are most comfortable to them and they’re skipping the things that are uncomfortable... all the buts are exactly the issues, they’re the problem. They’re not doing those things because that’s the problem for that organization. — Agile Coach

Therefore, while the long-term goal may be to improve multiple pain points, organizations should consider focusing on the largest perceived pain point first and adopt practices in the indicated practice category. Once practices from that category are adopted, and improvement is documented to some extent, then address the next pain point, and so on. However, as organizations’ agile deployments mature, or as the environment dictates, pain points and motivations for the adoption of agile methodologies may change. Organizations should continuously reevaluate the mix of agile practices used, to ensure that the focus of their agile practices is appropriate for their emerging context. By ensuring that the organizations’ motives for adopting agile methodologies are clearly understood by all, the type of practices on which agile method tailoring efforts are focused can be clarified.

Organizations: Identify key performance metrics going in

As we can see from Table 6, the concept of “fit” between the agile practices adopted and the organizational adoption motive(s) leading to agile method adoption might be a component of agile project success. In addition to determining their overall goal (i.e., motive) for utilizing agile methodologies, organizations might want to determine the performance factors on which they intend to focus the evaluation of their agile method deployment. While the connection of an agile practice and the performance outcome illustrated in Table 6 is rather consistent by motive, some performance factors may require different patterns of adoption. While a single category of agile practices (PM versus SDA) is associated with the majority of performance outcomes (e.g., the increased project success % for the improving software quality adoption motive is associated with a higher than average number of SDA-focused agile practices), in multiple cases, both categories of agile practices are connected to the desired impact (e.g., increased speed to completion for software quality is associated with a higher than average number of PM and SDA focused agile practices). We propose that organizations could identify the cell(s) in Table 6 on which they wish to focus (and then explore the potential adoption of the appropriately-focused agile practices).

Organizations: Clarify the motive for adopting agile methodologies, the performance metrics, and agile practice category focus

Although an organization’s motives may be focused, it is likely that various teams’ and team members’ goals may not be completely aligned with the organization’s goals (i.e., lack of fit). For instance, while an organization’s goals for agile

method adoption may be focused on anticipated efficiency gains, agile team members' goals may be learning to deliver software with new automated tools. Because the type of agile practice adopted (along with adoption motive) is linked to project performance outcomes, it is important for the organization to articulate why the adoption of agile methodologies is being attempted, which metrics will be used to evaluate agile project performance, and which agile practices are more likely to lead to success on the indicated metrics.

At the same time, it is likely that different teams will have different strengths and weaknesses. Organizations should focus on the category of practices (PM or SDA) to be adopted while leaving the particular mix of practices to be adopted more flexible. We found no evidence that *specific* PM-focused or SDA-focused practices led to performance, which supports the agile literature that often states that the agile practices are like tools in a tool box, ready to be applied when needed [39]. While a practice category (PM versus SDA) may be indicated, it is unclear as to whether, for instance, the use of retrospectives or iteration planning will be more useful in a particular context. For this reason, organizations should provide some leeway to teams in order to allow the teams to respond to their specific circumstances.

The way agile works is a group of people come together, there's somebody that clearly represents what needs to be done, the customer, and the team figures out how it's going to be done and they come up with the most effective way for them to work together given who they are and what it is they have to do. — Agile Coach

Teams: Adopt multiple practices, but choose wisely

Based upon our study, in order to achieve higher performance, teams might want to adopt a higher than average number of agile practices from the appropriate practice category (PM or

SDA). This means that dabbling in agile by adopting a single practice is not likely to lead to observable or sustainable increases in overall performance. Figure 2 suggests that significant overall project performance gains are achieved by the time an organization adopts just one of the agile practices under study here. While agile practitioners have always argued that agile practices build upon each other, and become more than the sum of their parts, the interplay between practices may cause a decrease in performance as more practices are adopted. As Figure 2 indicates, adopting multiple agile practices does not automatically translate into increased project success. Figure 3 provides a finer grained view of project success gains based on the category of agile practice adopted. Together, the graphs suggest that organizations should be patient as their teams adopt new configurations of practices, realizing that there may be an adjustment period. However, the benefits of implementing multiple practices seem to ultimately outweigh the issues related to the associated complexity.

Concluding remarks

Agile method adoption has evolved from a decision between particular agile methodologies into a more complex, adaptive process where the various agile practices are combined and recombined as necessary, being tailored to each organization's (and team's) needs and contexts. We have provided four guidelines for the tailoring of agile methodologies in various contexts, based on the findings presented regarding the relationship between the motives for adoption of agile methodologies, and the agile practices employed. Using the VersionOne State of Agile 2011 survey data and triangulating it with interviews with agile method practitioners, our study finds that three motives for agile adoption are each associated with different configurations of PM-focused and SDA-focused agile practices.

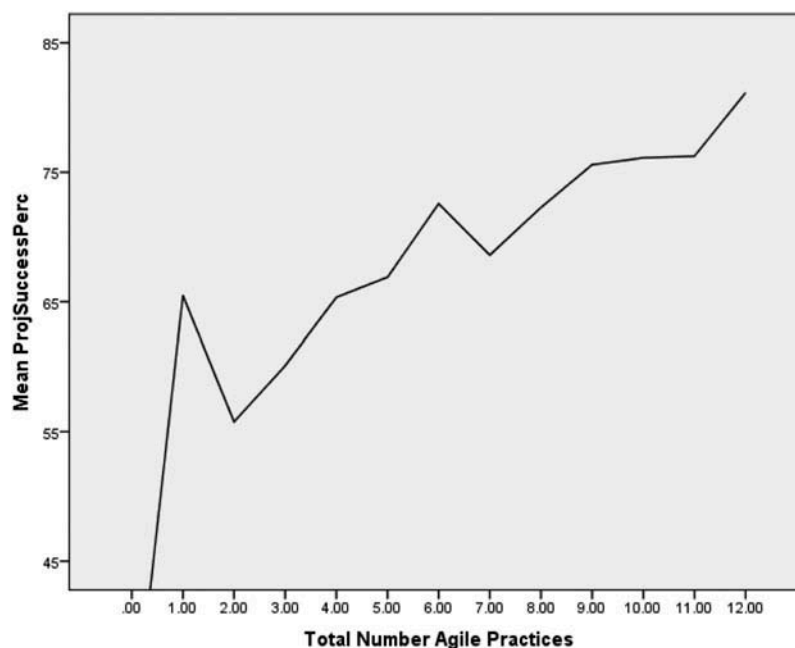


Figure 2. Total agile practices adopted vs. project success percentage.

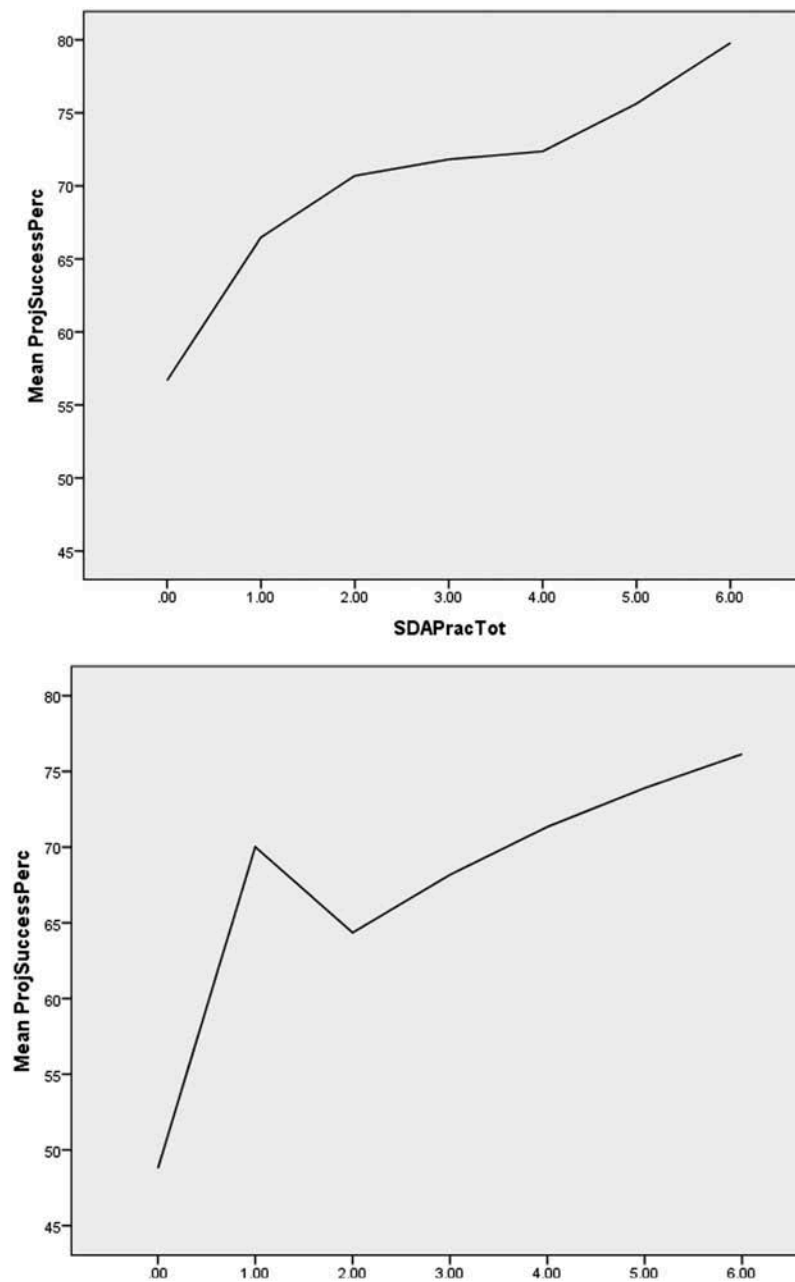


Figure 3. Top SDA and PM agile practices adopted vs. project success percentage.

From a practitioner perspective, the concept of “fit” with regard to aligning an organization’s motives for the adoption of agile methodologies and the agile practices utilized may provide practitioners the opportunity to explore an additional mechanism to aid the agile method tailoring process. Understanding an organization’s overall goal in terms of adopting agile methodologies (e.g., improve effectiveness) and which types of agile practices “fit” with the goal (e.g., iteration planning) may increase the success of the method tailoring process and, ultimately, agile software development projects.

References

- [1] Bass JM. Artefacts and agile method tailoring in large-scale offshore software development programmes. *Inf Softw Technol.* 2016;75:1–16.
- [2] Beck K. *Extreme programming explained: embrace change* (1st ed.). Boston (MA): Addison-Wesley Professional; 2000.
- [3] Begel A, Nagappan N. Usage and perceptions of agile software development in an industrial context: an exploratory study. *Proceedings of the First International Symposium on Empirical Software Engineering and Measurement*; 2007 September 20–21; Madrid, Spain. p. 255–264.
- [4] Berger H, Beynon-Davies P. The utility of rapid application development in large-scale, complex projects. *Inf Syst J.* 2009;19(6):549–570.
- [5] Campanelli AS, Parreiras FS. Agile methods tailoring: a systematic literature review. *J Syst Softw.* 2015;110:85–100.
- [6] Chow T, Cao D. A survey study of critical success factors in agile software projects. *J Syst Softw.* 2008;81(6):961–971.
- [7] Conboy K, Fitzgerald B. Method and developer characteristics for effective agile method tailoring: a study of XP expert opinion. *ACM Trans Softw Eng Methodol.* 2010;20(1):2:1–2:30.
- [8] Cram WA. Aligning organizational values in systems development projects. *Manage Res Rev.* 2012;35(8):709–726.

- [9] Denning S. The best-kept management secret in the world: agile. *Forbes*, 9 April 2012 [cited 2016 Feb 12]. Available from <http://www.forbes.com/sites/stevedenning/2012/04/09/the-best-kept-management-secret-on-the-planet-agile/#450365a42f47>
- [10] Dybå T, Dingsøyr T. Empirical studies of agile software development: a systematic review. *Inf Softw Technol*. 2008;50(9):833–859.
- [11] Fontana RM, Meyer V Jr, Reinehr S, Malucelli A. Progressive outcomes: a framework for maturing in agile software development. *J Syst Softw*. 2015;102:88–108.
- [12] Fruhling A, Vreede G. Field experiences with extreme programming: developing an emergency response system. *J Manage Inf Syst*. 2006;22(4):39–68.
- [13] Gandomani TJ, Nafchi MZ. An empirically-developed framework for agile transition and adoption: a grounded theory approach. *J Syst Softw*. 2015;107:204–219.
- [14] Gren L, Torkar R, Feldt R. The prospects of a quantitative measurement of agility: a validation study on an agile maturity model. *J Syst Softw*. 2015;107:38–49.
- [15] Hair JF, Anderson RE, Tatham RL, Black WC. *Multivariate data analysis with readings*. Englewood Cliffs (NJ): Prentice Hall; 1998.
- [16] Hajjdiab H, Taleb AS. Agile adoption experience: a case study in the UAE. *IEEE Second International Conference on Software Engineering and Service Science*; 2011 June 8–11; Beijing, China. p. 31–34.
- [17] Hanssen GK. Agile software product line engineering: enabling factors. *Softw Pract Exp*. 2011;41(8):883–897.
- [18] Heeager LT. Introducing agile practices in a documentation-driven software development practice: a case study. *J Inf Technol Case Appl Res*. 2012;14(1):3–24.
- [19] Holgado-Tello FP, Chacón-Moscoso S, Barbero-García I, Vila-Abad E. Polychoric versus Pearson correlations in exploratory and confirmatory factor analysis of ordinal variables. *Qual Quant*. 2010;44(1):153–166.
- [20] Keith M, Demirkan H, Goul M. Service-oriented methodology for systems development. *J Manage Inf Syst*. 2013;30(1):227–260.
- [21] Lawrence PR, Lorsch JW. *Organization and environment: managing differentiation and integration*. Boston (MA): Harvard University; 1967.
- [22] Layman L, Williams L, Cunningham L. Motivations and measurements in an agile case study. *J Syst Archit*. 2006;52(11):654–667.
- [23] Lee G, Xia W. Toward agile: an integrated analysis of quantitative and qualitative field data. *MIS Quart*. 2010;34(1):87–114.
- [24] McAvoy J, Butler T. Resisting the change to user stories: a trip to Abilene. *Int J Inf Syst Change Manage*. 2006;1(1):48–61.
- [25] Mangalaraj G, Mahapatra RK, Nerur S. Acceptance of software process innovations – the case of extreme programming. *Eur J Inf Syst*. 2009;18(4):344–354.
- [26] Maruping LM, Venkatesh V, Agarwal R. A control theory perspective on agile methodology use and changing user requirements. *Inf Syst Res*. 2009;20(3):377–399.
- [27] Misra SC, Kumar V, Kumar U. Identifying some important success factors in adopting agile software development practices. *J Syst Softw*. 2009;82(11):1869–1890.
- [28] Nerur S, Mahapatra RK, Mangalaraj G. Challenges of migrating to agile methodologies. *Commun ACM*. 2005;48(5):72–78.
- [29] Palmer SP, Felsing JM. *A practical guide to feature driven development*. Upper Saddle River (NJ): Prentice Hall; 2002.
- [30] Poppendieck M, Poppendieck T. *Lean software development: an agile toolkit*. Boston (MA): Addison-Wesley Professional; 2003.
- [31] Port D, Bui T. Simulating mixed agile and plan-based requirements prioritization strategies: proof-of-concept and practical implications. *Eur J Inf Syst*. 2009;18(4):317–331.
- [32] Ramesh B, Cao L, Baskerville R. Agile requirements engineering practices and challenges: an empirical study. *Inf Syst J*. 2010;20(5):449–480.
- [33] Schwaber K, Beedle M. *Agile software development with scrum*. Upper Saddle River (NJ): Prentice Hall; 2001.
- [34] Silva FS, Soares FSF, Peres AL, de Azevedo IM, Vasconcelos APLF, Kamei FK, Meira SRdL. Using CMMI together with agile software development: a systematic review. *Inf Softw Technol*. 2015;58:20–43.
- [35] VersionOne. 6th annual state of agile development survey. 2011 [cited 2013 Jan 23]. Available from www.versionone.com
- [36] Vidgen R, Wang X. Coevolving systems and the organization of agile software development. *Inf Syst Res*. 2009;20(3):355–376.
- [37] Vijayarathy L, Turk D. Drivers of agile software development use: dialectic interplay between benefits and hindrances. *Inf Softw Technol*. 2012;54(2):137–148.
- [38] Wang X, Conboy K, Pikkarainen M. Assimilation of agile practices in use. *Inf Syst J*. 2012;22(6):435–455.
- [39] West D, Grant T, Gerush M, D'Silva D. Agile development: mainstream adoption has changed agility. *Forrester Research*. 2010 [cited 2015 Apr 14]. Available from http://pmshow2012.programmedevelopment.com/public/uploads/files/forrester_agile_development_mainstream_adoption_has_changed_agility.pdf
- [40] Wood S, Michaelides G, Thomson C. Successful extreme programming: fidelity to the methodology or good teamworking? *Inf Softw Technol*. 2013;55(4):660–672.